

Module 6

Using Virtualization and Databases

Module Objectives

- Understand virtualization concepts.
- Understand containerization concepts.
- Differentiate relational and non-relational database types.

Lesson 6.1

Virtualization Concepts

Cloud+ CV0-004 Certification Exam Objective:

- 1.7 Compare and contrast virtualization concepts.

Virtualization Concepts

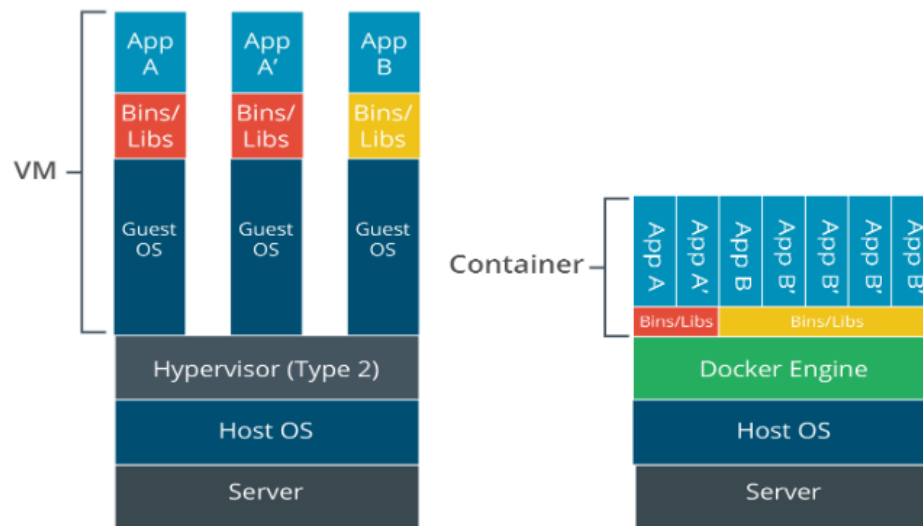
- Virtualization
- Stand-Alone Virtualization
- Deploy Virtual Machines in the Cloud
- Virtual Machine Configuration File Templates
- Clustering Virtualization
- Virtual Machine Cloning
- Network Types
- Types of Storage

Virtualization

- Partitions physical hardware into VMs
- VMs
- Admin responsibilities
- Hypervisor
- IaaS

Stand-Alone Virtualization

Container vs. VMs



Comparing virtual machines and containers. Note the virtual machine operating systems and hypervisor layer.

Deploy Virtual Machines in the Cloud

VM purpose choices

- General purpose
- Compute optimized
- Memory optimized
- Storage optimized

Storage type choices

- Premium solid-state drive (SSD)
- Standard SSD
- Standard hard disk drive (HDD)

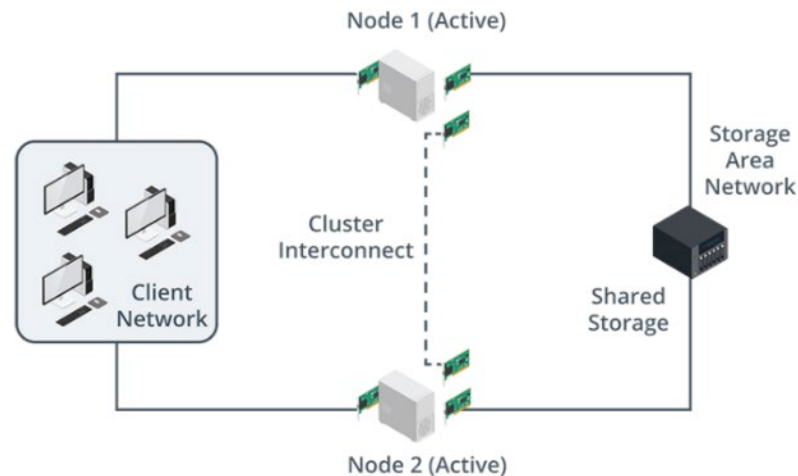
Virtual Machine Configuration File Templates

```
template.json (deleted) X
C:\Users\Administrator> AppData\Local\Temp> 1> Temp1_ExportedTemplate-515Support-Devlod38619419 (1):
24
32      "properties": {
33        "hardwareProfile": {
34          "vmSize": "Standard_D2s_v3"
35        },
36        "additionalCapabilities": {
37          "hibernationEnabled": false
38        },
39        "storageProfile": {
40          "imageReference": {
41            "publisher": "canonical",
42            "offer": "0001-com-ubuntu-server-focal",
43            "sku": "20_04-lts-gen2",
44            "version": "latest"
45          },
46          "osDisk": {
47            "osType": "Linux",
48            "name": "[concat(parameters('virtualMachines_Test_Server01_r
49            "createOption": "FromImage",
50            "caching": "ReadWrite",
51            "managedDisk": {
52              "storageAccountType": "StandardSSD_LRS",
53              "id": "[parameters('disks_Test_Server01_disk1_lccdc9a5fi
54            },
55            "deleteOption": "Delete",
56            "diskSizeGB": 30
57          }
58        }
59      }
60    }
61  }
```

Partial sample of a VM configuration file. Note the VM size, operating system (OS) information, and disk configurations. (Screenshot courtesy of Amazon.)

Clustering Virtualization

- Virtual machine clusters in the cloud
- Host affinity
- Hardware pass-through



A server cluster. (Images © 123RF.com)

Virtual Machine Cloning

- Cloning provides a snapshot of a VM's current configuration and data
- Options
 - Templates
 - Cloud providers

Network Types

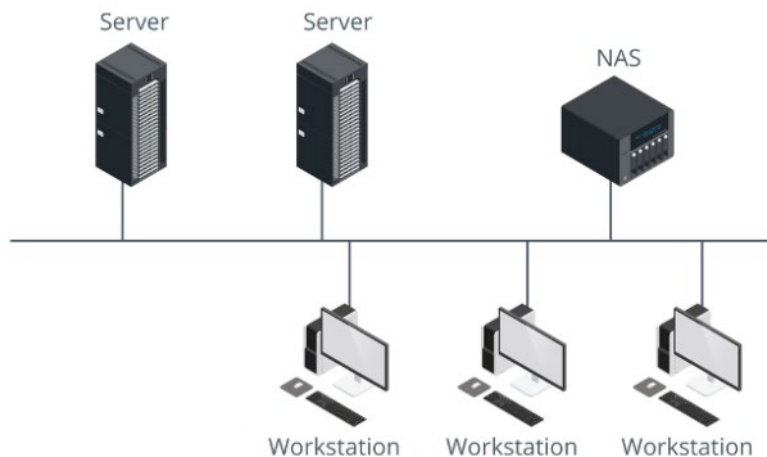
- Overlay networks
- Virtual machine networks
- Virtual machine network types

Types of Storage

- Local storage
- Network attached storage (NAS)
- Storage area network (SAN)

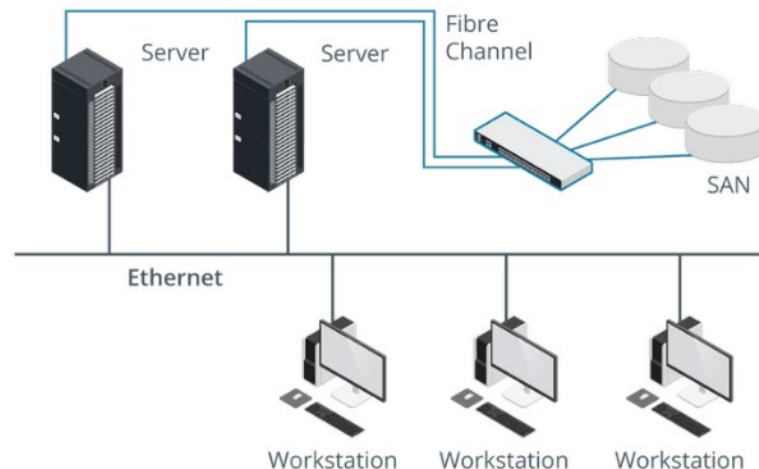
Types of Storage

NAS



Images 123RF.com

SAN



Images 123RF.com

Lesson 6.2

Containerization Concepts

Cloud+ CV0-004 Certification Exam Objective:

- 1.6 Compare and contrast containerization concepts.

Containerization Concepts

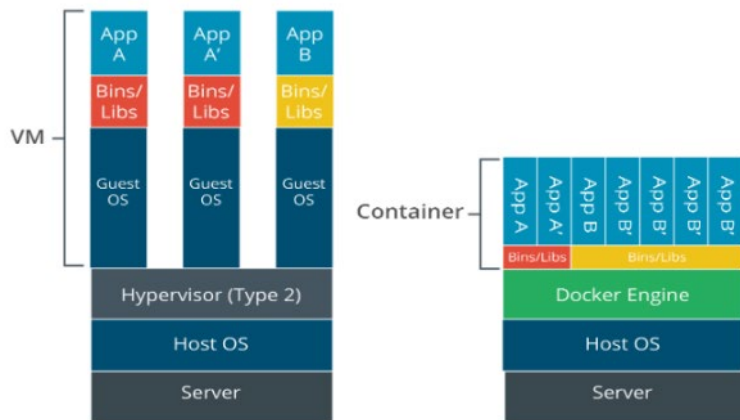
- Containerization
- Stand-Alone Containerization
- Deploy Containers
- Image Registries
- Containers in the Cloud
- Container Management

Containerization

- Software design
- Availability
- Scalability

Stand-Alone Containerization

Container vs. VMs



Comparing virtual machines and containers. Note the container engine with multiple containers.

- Configuration file
- Container image
- Running container

Deploy Containers

Docker Subcommands

Subcommand	Description
info	System-wide info
port	List port mappings for the specified container
pull	Pull an image from a registry
push	Upload an image to a registry
ps	List containers
rm	Remove containers
rmi	Remove images
run	Run a command in a container

Deploy Containers

Docker Container Subcommands

Subcommand	Description
start	Start a stopped container
stop	Stop a started container
restart	Restart a container
attach	Attach local STDIN, STDOUT, STDERR to a container
inspect	Display detailed container information
logs	Display container logs
ls	List existing containers

Image Registries

- Storage repositories for container images used by applications and developers
- Essential for DevOps and CI/CD processes
- Public (e.g., Docker Hub) and private registries
- Cloud provider registries
 - Amazon Elastic Container Registry (ECR)
 - Azure Container Registry
 - Google Artifact Registry

Containers in the Cloud

- Benefits
 - Scalability
 - Metered services
 - High availability
 - Reduced management overhead
- Cloud Container Services
 - AWS: Amazon Elastic Container Service
 - Azure: Azure Kubernetes Service
 - Google: Google Compute Engine

Container Management

- Port mapping
- Storage types
 - Ephemeral storage
 - Persistent storage
- Orchestration

Lesson 6.3

Database Concepts

Cloud+ CV0-004 Certification Exam Objective:

- 1.9 Explain the importance of database concepts.

Database Concepts

- Database Types
- Relational Databases
- Non-Relational Databases
- Database Deployment Options
- Migrate Databases

Database Types

- Relational
- Non-relational

Relational Databases

- Data organization
 - Column: Data attributes
 - Row: Instances of that data
- Structured Query Language (SQL)

Non-Relational Databases

- Document-oriented
- Key-value
- Wide column
- Graph stores

Database Deployment Options

	Self-managed Databases	Provider-managed Databases (DaaS)
Advantages	• Control of data • Flexibility • Avoids vendor lock-in • Control over database choices	• Structured • Maintenance-free/support provided • Automation • Probable cost savings and pay-as-you-go cost model • Built-in scalability
Disadvantages	• Significant investment in time and resources • Technical skills • Commitment	• Possible vendor lock-in • Security concerns in public clouds • Fewer customization options • Data sovereignty issues

Migrate Databases

- Direct migrations
- Cross-service migrations

Module Summary

- Virtualization: Essential for cloud solutions; managed daily to meet cost, performance, and feature needs
- Databases: Hosted on VMs; crucial for storing and retrieving business information
- Types: Relational and non-relational databases
- Role: Cloud admins help choose the best database solutions